

in sustained transfer writes but for bursty transfers, it should not be a problem at all.

Right next to the controller is the ASMedia ASM2362 bridge which is what allows the USB Type-C port to talk to the controller. The ASM2362 is a first-generation PCI Express bridge chip with speeds up to 1000 MBps and is compliant with NVMe 1.2.1 revision. It's got a USB 3.2 Gen2 upstream interface and a PCIe Gen3x2 downstream interface. It's yet another extremely popular bridge used by a lot of external SSDs and SSD enclosures

SOFTWARE

The drive comes with the Samsung Magician software on it. You can choose to install the software and make use of the additional monitoring and security features or just ignore it and use the drive normally. Checking the Samsung Magician software every now and then, for firmware updates is essential since you might come across an update that genuinely improves the life or performance of the drive (to some extent). It will help you avoid the infamous Samsung 980 Pro SSD failure bug

about 1067 MBps for Sequential 1 MB read operations and for write operations that drops by barely 10-15 MBps.

Switching to Sequential 2 MB read operations drops the write speed to about 800 MBps and the read speeds also come down to about 770 MBps.

To simulate an extended transfer we also ran a batch file to repeatedly copy and delete a massive 20 GB file to see if the controller would throttle once it got sufficiently hot. Turns out, the massive pink thermal pad does what it was designed to, wonderfully. We didn't notice a drop in speeds even after 20 minutes of heavy usage.



Boo-Man's thoughts

"The Samsung T7 Shield, a portable SSD, might have some durability, but let's be real, it's already outdated in the fast-paced tech world, and the competition is miles ahead. I guess you could use it to store your nostalgia for older tech!"

such as the ones you'd find on Amazon by the dozen.

Lastly, there's the NAND chips where all your precious data gets stored. The ones we saw were 128-layer TLC Samsung K9DVB8J1B-DCK0 NAND chips which are part of the 6th Gen V-NAND lineup that was launched in August 2019. These chips can have up to 136 layers with over 100 active layers. Compared to the 5th Gen V-NAND chips, these are 10 per cent faster and also consume 15 per cent power amount of power. Each chip used here gives the drive 1 TB of capacity.

Overall, we see a pretty standard design on the Samsung T7 Shield with nothing out of the ordinary.

which would sometimes render the 980 Pro stuck in read-only mode and therefore unusable. Always, read up about the firmware before updating.

PERFORMANCE

The Samsung T7 shield unit that we received was a slightly used one and just by the looks of the packaging had seen better days. Fortunately, one of our colleagues had recently purchased the same drive so we had a reference unit to cross-check our performance numbers.

We plugged the unit into our test rig and started with ATTO. Around the 128K file transfers, we see a consistent throughput of around 980-1000 MBps. CrystalDiskMark also sees

VERDICT

Samsung has, no doubt, a good external SSD on its hands. The Samsung T7 Shield is a consistent performer and locks horns with its competitors across a lot of scenarios. However, it is a little perplexing for Samsung to be a little behind the times.

Especially considering that Samsung is one of the major manufacturers which makes the chips that are used by a lot of the other external storage vendors. Its major competitors had 20 Gbps drives years before the T7 came out. When you are thinking of a 1 TB solution then Samsung, WD, SanDisk, Crucial and Seagate are all on the same page.

You get the drives for a little over 7K and they all come with a 3-year warranty coverage. Moving to 2 TB and higher capacities makes the SanDisk Extreme Pro the defacto recommendation because it comes with a 5-year warranty coverage. And if you're looking to get 20 Gbps drives, then the T7 doesn't even make the cut.

-Mithun Mohandas

SPECIFICATIONS

Capacity: 4 TB | NAND: 6th Gen V-NAND | Rated speed: 1000 MBps | Interface: USB 3.2 Gen2 | Weight - 98 grams | Dimensions (LxWxD): 88 x 59 x 13 mm | Warranty: 3 years.

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